

Instructions for Thesis Preparation of Master Program (Parts of master thesis, in the order they should appear)

The best way of writing master thesis is to prepare an extended outline. Supervisor should carefully review the outline and check whether there are any unnecessary materials that are not directly related to the problem statement. It is expected that the master thesis is done with a defined research question that has been solved through well designed research work and the results have been presented systematically. Writing the thesis requires complete organization of the arguments and results. Choose section titles and wordings to clearly give the idea about the findings of the study. This guideline address the major issues related to the structure and style of master thesis.

The fundamental element of master thesis generally comprises the following:

1. **Title:** The wording of the title should be simply a characterization of the theme. This may contain major elements the candidate will address. Preferably titles should be simple and direct, without a lot of qualifying terms or phrases. It is better to solicit the opinion of the research supervisor. Any abbreviation in the thesis title should be avoided.
2. **Approval sheet:** This part should be followed as per proforma shown.
3. **Table of contents:** This should list in sequence, with page number, all major subdivision of the dissertation such as chapters, the references, appendices etc.
4. **Declaration:** I declare that the thesis hereby submitted by me for the MS degree at the Patuakhali Science and Technology University is my own independent work and has not previously been submitted by me at another university/faculty for any degree. This declaration must be signed by the candidate with date of submission of the thesis. This thesis is an independent compilation and attachment to your thesis. Work without this thesis will not be accepted. The name and signature of the candidate will appear in all hard binding copies of the thesis. Declaration page should not be given in soft binding copies. Candidates are strongly advised to check the regulations for specific requirements for the master degree.
5. **Acknowledgements:** Any help received from any person or institution, in the provision of facilities, should be formally acknowledged. The full name and position of all persons acknowledged are essential.
6. **Abstract:** This is the only section of the thesis that is read by most of the readers. The abstract extracts the key points of the thesis, highlighting its purpose, methods, major findings and brief conclusions. It does not include figures, table or citations. If more than one hypothesis is tested this should be stated in the abstract.

The abstract page contains: (i) Title of the Thesis; (ii) Full name of the author; (iii) The word ABSTRACT in bold uppercase just before the text; (iv) The text should be single paragraph in between 250-300 words, in single line spacing.

7. **Introduction:** This section provides the background for the research work and focused on the topic investigated. The purpose is to convince the reader of the importance of the work done. It includes a historical basis for selecting the problem and shade light on some point of missing pieces of information that will prove pertinent to the problem studied. Brief background information is necessary to make the readers fully familiar to the problem addressed in the thesis.
8. **Objectives:** This can be included with the introduction or presented as a separate section. The objectives should be very specific and quantifiable. Objectives should not be confused with the ultimate goal of the work.
9. **Review of literature:** A separate section is necessary to cover the amount of published literature related to the topics studied. The review should be comprehensive but not too elaborative. It is not necessary to cite every study even remotely connected to the subject at hand. It is essential to cite those that are most important for the work done. The reference should be current, with the exception of historical, original work done. Ideally source should be primary ones, representing the relevant work done the field to date and relevant to the specific area of study. Sources whose theories or opinions conflict with the study should also be included. All information is to be appropriately cited, as plagiarism (it is define as presenting someone else's work as one's own) is illegal. It may require some discussion with the research supervisor to sort out the relative importance of the articles. Critically evaluate the articles and make some independent assessment of their importance to the research question that has been done for the thesis. Traditional "name and year" system should be followed for citation in the text.
10. **Materials and Methods/Methodology:** Once the research experiments/studies are thoroughly planned, presenting the methodology can be the most straightforward part of the thesis work. The materials, methods and experimental designs should be described in a way that they can be repeated by other workers. This chapter may have one or several sections and subsections that can vary with the nature of the experiments. For instance, in case of the field experiment, the following information should be provided: (i) Site characteristics: geography, climate, soil properties (ii) Design of the experiments: Randomized Block Design, Split-plot Design, and number of replicates (iii) Agronomic practice: Plant materials, fertilization, irrigation, treatments (iv) Observation and Sampling Methods: phonological stages, light measurements, root sampling method, harvesting.
Precise description of the devices and methods including the citation are required as well. Appropriate data analysis procedures need to be written considering the hypotheses tested. Statistical methods should be described with sufficient clarity to allow the reader to verify the reported results. The appropriate tests together with the software used for the analysis should be cited.
11. **Results and Discussion:** This section includes analyses of data and presentation of results. Before writing this section, the writer should organize his/her data in tables,

graphs, photographs etc. Various alternatives should be tried to find out the best. Once the data are organized into tables and other illustrations, the textual part of the results can now be written quite easily. Results should be presented logically rather than chronologically, using sub-headings and paragraphs for observations of different experiments. Tables and figures should be presented with clear, concise, self-evident captions and labels. The data presented in the tables should not be repeated in the text, rather the salient features and trends of the data presented the table should be indicated in the text. Statistical analysis should indicate the relationships between variables. Test statistics, level of significance, degrees of freedom and sample size need to be provided in brackets after each data analysis result.

Discussion should provide clear interpretation of results in the context of present state of knowledge. Writing a good discussion requires a lot of thinking. The writer must remember few basic points while writing discussion. A writer cannot discuss any data that have not been presented in the results; neither can he include any new data in the discussion section. Moreover, one should not reconsider every scrap of his/her results in the discussion. The thumb rule is “do not discuss if no discussion is needed”. An ideal discussion should have the following components: Key message summarizing the major achievement, necessary interpretation of the findings, comparisons of the results with relevant findings of others to show how this result fits in the current state of the knowledge, and necessary explanation for any deviation. An honest writer would qualify his/her results in the discussion section by mentioning any limitations and shortcomings of his/her experiments.

- 12. Conclusions:** It is not a rambling summary of the thesis: they are short, concise statements of the inferences that have been made out of the study. It helps to organize these as short numbered paragraphs, ordered, from most to least important. All conclusions should be directly related to selective objectives.

Future direction of research should be included, so that researchers can pick up the ideas that have been generated while this work was done.

- 13. References:** Most examiners scan the list of references looking for the important works in the field, inclusive of their own if applicable. Make sure they are listed and referred to in section of review of literature. All references given must be listed in alphabetical order by author’s surname. Works by the same author’s should be listed in chronological order.

A. Citing References in the Text

Type of citation	Narrative citation in text	Parenthetical citation in text
One author	Walker (2007)	(Walker, 2007)
Two authors	Walker and Allen (2004)	(Walker & Allen, 2004)
Three and more authors	Wasserstein et al. (2005)	(Wasserstein et al., 2005)
Groups	National Institute of Mental Health (NIMH, 2003)	(National Institute of Mental Health [NIMH], 2003)
Reference with single page number/chapter,	Hedetoft (2013)	(Hedetoft, 2013)

etc		
Reference with page range	Haan (2005, pp. 36-37)	(Haan, 2005, pp. 36-37)
Same author, two works from different years	Todres (2011, 2012)	(Todres, 2011, 2012)
Same author, two works from same year	Derryberry and Reed (2005a, 2005b)	(Derryberry & Reed, 2005a, 2005b)

Important Notes:

1. Each reference cited in text must appear in the reference list and vice-versa.
2. When citing two or more works together, arrange the in-text citations alphabetically in the same order in which they appear in the reference list:

Recent research in developmental psychology has explored this issue (Bergstrom et al., 2006; Clément, 2010; Harris & Corriveau, 2011; Harris & Koenig, 2006; Heyman, 2008; Heyman & Legare, in press; Koenig & Harris, 2005).
3. If you want to include several different citations in one set of brackets, they should be in alphabetical order, separated by a semi-colon:
(Durie, 2003; McShane & Travaglione, 2007; Ministry of Education, 1996).
4. To cite more than one work by the same author(s), list the surname of the author(s) and then each different year of publication, separated by a comma:
(Ministry of Education, 1996, 1999).

B. The Reference List

Type of Reference	Format	Examples
Journal Article (1 author)	Lastname, A. (year). Title of the article in sentence case. <i>Journal in Title Case</i> , Volume(Issue), Firstpage-Lastpage.doi	Paivio, A. (1975). Perceptual comparisons through the mind's eye. <i>Memory & Cognition</i> , 3(1), 635–647.
	Attention: The issue number is not compulsory.	Jerrentrup, A. (2018). Teaching medicine with the help of “Dr. House.” <i>PLoS ONE</i> , 13(3), Article e0193972.
	If the journal article has an article number instead of a page range, include the word “Article” and then the article number instead of the page range.	Andreff, W. (2000). The evolving European model of professional sports finance. <i>Journal of Sports Economics</i> , 1, 257–276. https://doi.org/10.1177/152700250000100304
	Journal articles (but also other types of references, such as books) can have a doi or url at the end, but that is not compulsory.	

Journal Article (2-7 authors)	Lastname, A., & Lastname, B. (year). Title of the article in sentence case. <i>Journal in Title Case, Volume(Issue)</i> , Firstpage-Lastpage. doi	Becker, L. J., & Seligman, C. (1981). Welcome to the energy crisis. <i>Journal of Social Issues</i> , 37(2), 1–7.
		Cheryan, S., & Meltzoff, A. N. (2015). Cultural stereotypes as gatekeepers. <i>Frontiers in Psychology</i> , 6, Article e0214466. https://doi.org/10.3389/fpsyg.2015.00049
		Frischlich, L., Hahn, L., & Rieger, D. (2021). The promises and pitfalls of inspirational media: What do we know, and where do we go from here? <i>Media and Communication</i> , 9(2), 162–166. https://doi.org/10.17645/mac.v9i2.4271
Journal Article (more than 7 authors)	Lastname, M., Lastname, N., Lastname, O., Lastname, P., Lastname, Q., Lastname, R., Lastname, M., . . . Lastname, Z. (year). Title of the article in sentence case. <i>Journal in Title Case, Volume(Issue)</i> , Firstpage-Lastpage. doi	Gonzalez, W. L., Hughes, W., Carter, D., Campbell, C., Baker, A. B., Flores, T., Gray, W. E., . . . Lee, L. H. (2018). National character does not reflect mean personality trait levels in 49 cultures. <i>Science</i> , 3(10), 96–100.
Book	Lastname, A. (year). <i>Title of the book in sentence case</i> (edition, Volume). Publisher. doi Attention: The edition, volume, and DOI are not compulsory.	Strunk, W., Becker, E., & White, E. B. (1979). <i>The guide to everything and then some more stuff</i> (3rd ed.) Macmillan.
		Lippe, T. L. (Ed.). (2016). <i>Student affairs for academic administrators</i> . Stylus Publishing; ACPA Books.
		Kearney, D. J., & Simpson, T. L. (2020). <i>Concise guides on trauma care. Mindfulness-based interventions for trauma and its consequences</i> . American Psychological Association. https://doi.org/10.1037/0000154-000 .
Book Chapter	Lastname, A. (year). Title of the chapter in sentence case. In B. Lastname, C. Lastname, & D. Lastname (Eds.), <i>Title of the book in sentence case</i> (Volume, pp. firstpage-lastpage). Publisher. doi	Bergquist, J. M. (1992). German Americans. In J. D. Buenker & L. A. Ratner (Eds.), <i>Multiculturalism in the United States: A comparative guide to acculturation and ethnicity</i> (Vol. 3, pp. 53–76). Greenwood.
		Baker, F. M., & Lightfoot, O. B. (1993). Psychiatric care of ethnic elders. In A. C. Gaw (Ed.), <i>Culture, ethnicity, and mental illness</i> (pp. 517-552). American

		Psychiatric Press.
		Scott, D. (2005). Colonial governmentality. In J. X. Inda (Ed.), <i>Anthropologies of modernity</i> (pp. 21–49). Retrieved from http://www3.interscience.wiley.com/cgi-bin/bookhome/117909832
Newspaper Article / Newsletter (in paper)	Lastname, A. (year, month day). Title of the article in sentence case. <i>Title of the Newspaper/Newsletter in Title Case</i> , Firstpage-Lastpage.	Schwartz, J. (1993, September 30). Obesity affects economic, social status. <i>The Washington Post</i> , 1–4.
Newspaper Article / Newsletter (online)	Lastname, A. (year, month day). Title of the article in sentence case. <i>Title of the Newspaper/Newsletter in Title Case</i> . www.website.com	Carey, B. (2019, March 22). Can we get better at forgetting? <i>The New York Times</i> . https://www.nytimes.com/2019/03/22/health/memory-forgetting-psychology.html
Website	Lastname, A. (year). <i>Title of the website article in sentence case</i> . Name of the Website in Title Case. www.website.com	Cain, K. (2012). <i>The negative effects of Facebook on communication</i> . Social Media Today. http://socialmediatoday.com
Published Thesis or Dissertation	Lastname, A. A. (year). <i>Title of thesis in sentence case</i> [Doctoral or Master's thesis or dissertation, Name of the Institution]. Repository. www.website.com	Kelly, C. B. D. (2018). <i>The art of coffee roasting</i> [Doctoral thesis, The University of Waikato]. The University of Waikato Research Commons. https://hdl.handle.net/10289/11614 .
Unpublished Thesis or Dissertation	Lastname, A. A. (year). <i>Title of thesis in sentence case</i> [Unpublished doctoral or Master's thesis or dissertation]. University.	Imber, A. (2003). <i>Applicant reactions to graduate recruitment and selection</i> [Unpublished doctoral dissertation]. Curtin University.
Conference paper, keynote address, poster presentation, etc. (unpublished)	Lastname, A. (Year, Month Day-Day). <i>Title of paper in sentence case</i> [Type of material]. Name of Conference, City, Country.	Whipple, S. (2018, March 6-9). <i>Control beliefs as a moderator of stress on anxiety</i> [Paper presentation]. Southeastern Psychological Association 64th Annual Meeting, Charleston, SC, United States.
Conference paper in Proceedings (published as a book)	Lastname, A. B. (Year). Title of paper. In A. Lastname (Ed.; if applicable), <i>Proceedings book title in sentence case</i> (pp. firstpage-lastpage). Publisher.	Cismas, S. C. (2010). Educating academic writing skills in engineering. In P. Dondon & O. Martin (Eds.), <i>Latest trends on engineering education</i> (pp. 225-247). WSEAS Press.
Comment on Facebook	Author of the comment, A. (Year). First sentence of the comment [Facebook comment on the Month	Wangail, T. (2017). Since Kashmir mass uprising, all Wimax connections were suspended almost for more than three

Post	Day, Year, post by Author of the Post]. www.website.com	months [Facebook comment on the January 23, 2017, post by N Rigzin]. https://www.facebook.com/groups/904515826309195/permalink/1291073837653390/?comment_id=1293505600743547
Brochure	Author. (Year). <i>Title of document</i> [Brochure]. Publisher.	Ministry of the Environment. (2007). <i>Conserving water in Ontario</i> [Brochure]. Ministry of the Environment.

Notes:

This report is prepared based on APA style of citation and reference.

APA style requires a hanging indent, which means that the first line of each reference entry is aligned flush with the left margin and each subsequent line has a hanging indent of 1.27 cm or 0.5 inch.

- 14. Appendices:** Any material which increases the main text volume, but is important to justify the results of a thesis should be appended. The material, which is either too detailed or not that central, should be included in the appendix. Examples includes immense tables of data, program listings, detailed maps etc.
- 15. Page layout and paper:** The text and wherever possible, all the material of the thesis, including illustrations should be produced on A4-size using white paper of weight 80g/m² and printed in black ink. All margins should be 1.0 inches wide, except the left margin which should be 1.5 inches wide to allow for binding.
- 16. Typing and fonts:** The Times New Roman font should be used for the thesis text, headings and figure captions. A 12 pt font should be used for the main text. A 10 pt font should be used for the footnote. Although a slight larger 14 pt font may be used for headings and sub-headings, respectively. A 16 pt font may be used for title. Headings may be in a bold font. All submitted theses copies should be identical to one another.
- 17. Paragraph formatting:** One-and-a-half line spacing should be used throughout the text with the following exceptions; which should be single-line spacing: (i) the table of contents; (ii) tables; (iii) footnotes and endnotes; (iv) long quotations of more than 40 words; (v) computer program in listings/codes. A 6 pt extra space should be provided after each paragraph and each reference using paragraph formatting features of word processor.
- 18. Numbering:** The chapter headings begin at level 1 and numbered sequentially as 1, 2, 3 etc. The level 2 headings (sub-headings) should be numbered as 1.1, 1.2, 1.3 etc. while level 3 headings should be numbered as 1.1.1, 1.1.2, 1.1.3 etc. If more than one appendix is included, these should be numbered separately as Appendix 1, Appendix 2 etc. Figure should be numbered as Figure 1, Figure 2 etc. without distinguishing between different kinds of figure. Tables within the text should be numbered consecutively in a single sequence Table 1, Table 2.
- 19. Pagination:** The title page, approval pages, dedication, acknowledgements, abstract and contents pages etc. should be numbered with Roman numerals in lower case as i,

iii, iv, etc. without showing number on the first page. The rest of the thesis should be numbered in a single sequence in Arabic numbers as 1, 2, 3, 4, 5 etc. starting again with 1 on the first page of the main text (Introduction). Illustration, charts, tables, diagrams etc., placed within the text of the thesis should be paginated as if they are pages of text. Page number should be in the bottom of the page in the centre, midway between the bottom edge of the paper and the last line of text on the page.

- 20. Submission of non-paper media:** One soft copy through e-mail along with the hard copy of the dissertation should be submitted to the Chairman of the relevant department and Central Library of PSTU. Where appropriate to the academic work being conducted, this additional soft copy may be useful. Electronic media must be unalterable and that soft copy submissions should normally be treated as supplementary to the paper dissertation.

(Specimen showing the layout of the cover page hard-binding for Master Thesis)

**CRITICAL PERIOD OF WEED COMPETITION IN
TRANSPLANT AUS RICE cv. BRRI dhan27 UNDER
NON-SALINE AGRO-ECOSYSTEM**

..... Thesis
Full name of the student

Department of
Patuakhali Science and Technology University
Patuakhali

June / December Year

N.B.: Hard-binding cover page should be in royal blue rexin and typescripts on the cover should be in golden color.

(Specimen page showing the layout of inner title page for hard-binding of Master Thesis)

**CRITICAL PERIOD OF WEED COMPETITION IN
TRANSPLANT AUS RICE cv. BRRI dhan27 UNDER
NON-SALINE AGRO-ECOSYSTEM**

A Thesis

Submitted to
Patuakhali Science and Technology University, Patuakhali
In Partial Fulfillment of the Requirements
for the Degree of Master of
in
.....

By
Student's Name
Roll/ID No.:
Registration No.:

Department of
Patuakhali Science and Technology University
Patuakhali

June / December Year

(Specimen page showing the layout of an approval page for hard-binding of Master Thesis)

**CRITICAL PERIOD OF WEED COMPETITION IN
TRANSPLANT AUS RICE cv. BRRI dhan27 UNDER
NON-SALINE AGRO-ECOSYSTEM**

A Thesis

Submitted to
Patuakhali Science and Technology University, Patuakhali
In Partial Fulfillment of the Requirements
for the Degree of Master of
in
.....

By
Student's Name

Approved as to style and contents by

Signature
(Name)
Supervisor

Signature
(Name)
Co-Supervisor

Signature
(Name)
Chairman, Defense Committee
and
Chairman, Department of

June / December Year

DECLARATION

I declare that, the thesis hereby submitted by me for the Master degree at the Patuakhali Science and Technology University is my own independent work and has not previously been submitted by me at another university/faculty for any degree.

Date:

Signature
.....
(Full name of the student)

N.B.: The name and signature of the candidate will appear in all hard binding copies of the thesis. Declaration page should not be given in soft binding copies.

Table of Contents

	Declaration	iii
	Acknowledgements*	iv
	Abstract	v
1.	Introduction	1
2.	Review of Literature	
3.	Materials and Methods	
4.	Results and Discussion	
5.	Conclusions	
6.	References	
7.	Appendices (if any)	

*This is not allowed to write in the soft binding copies of the dissertation.

Professor Dr. Md. Fazlul Hoque
Dean
Postgraduate Studies